THE IRREDUCIBLE COMPONENTS OF A FINITE IRREDUNDANT UNION ARE EXACTLY ITS MEMBERS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that every irreducible subset is contained in an irreducible component. Invoked inside the counted proof (source line 1156). Finds a component containing one of the given closed subsets.

Source reference. Original lemma lemma-irreducible: sources/stacks-topology/pinned-original-source.tex, source lines 1103–1103. Navigation pointer only: the counted text refers to the Stacks item lemma-irreducible, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space and suppose $X = \bigcup_{i=1,\ldots,n} X_i$ where each X_i is an irreducible closed subset of X and no X_i is contained in the union of the other members. Then each X_i is an irreducible component of X and each irreducible component of X is one of the X_i.*

Proof. Let Y be an irreducible component of X. Write $Y = \bigcup_{i=1,\ldots,n}(Y \cap X_i)$ and note that each $Y \cap X_i$ is closed in Y since X_i is closed in X. By irreducibility of Y we see that Y is equal to one of the $Y \cap X_i$, i.e., $Y \subset X_i$. By maximality of irreducible components we get $Y = X_i$.

Conversely, take one of the X_i. By Lemma lemma-irreducible it is contained in an irreducible component Y of X. By the previous paragraph $Y = X_j$ for some j. So $X_i \subset X_j$ and since the original union does not have redundant members, $X_i = X_j$, which is an irreducible component. □

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